

BI Definition

- Howard Dresner (1989) - founder and CRO of Dresner Advisory Services
- Dresner used the phrase to describe a set of concepts and methods to improve business decision-making by utilizing fact-based support systems. (Khan A. (2025) *Business Intelligence and Data Analysis in the Age of AI*, Mercury Learning and Information)

Howard Dresner

- Dresner's early definition laid the groundwork for the field of business intelligence, which has since evolved into a crucial component of modern business strategies, involving data analysis, reporting, and the use of technology to provide actionable insights for organizations.

Business Intelligence Overview

- The primary objective of Business Intelligence (BI) is to empower organizations with actionable insights derived from their data, fostering informed decision-making and strategic planning. By systematically collecting, processing, and analyzing data, BI enables enterprises to gain a comprehensive understanding of their internal processes, market dynamics, and customer behavior.

Evolution of Business Intelligence

- Beginning in the late twentieth century, BI emerged as a concept for improving decision-making processes by leveraging data-driven insights.
- Initially focused on reporting and querying, BI evolved with the advent of data warehousing, extract, transform, load (ETL) processes, and multidimensional analysis.
- In the 2000s, a shift toward more user-friendly interfaces and interactive dashboards occurred, making BI accessible to a broader audience.
- The rise of big data and cloud computing in the following decade further propelled BI, enabling organizations to handle larger datasets and facilitate real-time analytics.
- Today, BI incorporates advanced analytics, machine learning, and Artificial Intelligence, providing predictive and prescriptive capabilities for more proactive decision-making.
- The evolution of BI reflects a continuous adaptation to technological advancements, expanding its capabilities and impact on organizational strategies.

How Business Intelligence (BI) Is used:

Scenario: A Retail Company Improving Sales and Inventory Management

1.Data Collection - The company gathers data from multiple sources:

- Point-of-sale (POS) systems (daily sales)
- Online store transactions
- Inventory records
- Customer loyalty programs

2.Data Analysis - Using BI tools, the company analyzes:

- Which products sell the most by location and season
- Peak shopping days and hours
- Customer buying patterns (e.g., repeat customers, popular product bundles)

4.Dashboards and Reports - Managers view interactive dashboards that show:

- Real-time sales performance
- Low-stock and overstocked items
- Monthly and weekly sales trends

5.Decision-Making - Based on BI insights:

- The company restocks fast-selling items before they run out
- Slow-moving products are discounted or discontinued
- Promotions are targeted to specific customer groups

6.Business Impact

- Reduced inventory costs
- Increased sales and customer satisfaction
- Faster, data-driven decisions instead of guesswork

Role of data-driven decision-making in modern enterprises

Business Intelligence applications play an extremely important role in this process.

BI Application Categories

- **Operations Reporting** - This serves as a foundational layer for organizations to understand and optimize their day-to-day activities. It uses online analytical processing and visualization.
- **Analytics** - encompassing a range of techniques, including predictive analysis and statistical analysis.

Enter a growing field of BI Professionals

- According to the Philippine Skills Framework Companies report the largest skills gap in data analytics, and the demand for data analytics skills is growing over 15 times faster than the demand for the average skill in the US. (*PSF, 2025*).
- Every business process, in organizations of all types and sizes, generates huge volumes of data. This data may be stored in different locations and formats that are difficult to access and don't provide useful insights.
- To use that data to effectively inform business decisions, it needs to be retrieved, structured, interpreted, and reported in usable formats that enable stakeholders to understand and act.

Global In-House Center Career Map

Business Intelligence Tools

Business intelligence tools form a diverse ecosystem, each category catering to distinct facets of data analysis and reporting.

Data Visualization Tools

- Tableau by Salesforce
- Power BI by Microsoft

Reporting Tools

- SAP
- Crystal Reports
- IBM Cognos Reporting
- Designed to create interactive visual representations facilitating a comprehensive understanding of complex datasets
- Focus on structuring and presenting data in a clear and organized format, which is crucial for conveying critical business information effectively.